AI 3000/7010: Reinforcement Learning
Tutorial 5 Solutions

1 Higher Order Markov Policies

Consider a first-order MDP $(\mathcal{S}, \mathcal{A}, P, \mathcal{R})$.

(a) **Second-order policy.**

Suppose that the policy depends on the last two states:

$$\mathbb{P}^\pi(A_n = a \mid S_n = s, S_{n-1} = \tilde{s}, A_{n-1} = \tilde{a}, \dots, S_0 = s_0, A_0 = a_0) = \pi(a \mid s, \tilde{s}).$$

Prove that $\{S_n\}_{n\geq 0}$ process is a second-order Markov chain. That is, show that

$$\mathbb{P}(S_{n+1} = s' \mid S_n = s, S_{n-1} = \tilde{s}, \dots S_0 = s_0) = P^\pi(s' \mid s, \tilde{s})$$

for an appropriate transition kernel P^π, and explicitly derive this kernel.

Solution. Let

$$(H_n = h_n) \equiv (S_n = s, S_{n-1} = \tilde{s}, \dots, S_0 = s_0).$$

Conditioning on the action selected at time n and using the first-order transition property of the MDP gives

$$\begin{aligned}
\mathbb{P}(S_{n+1} = s' \mid H_n = h_n) &= \sum_{a\in\mathcal{A}} \mathbb{P}(S_{n+1} = s', A_n = a \mid H_n = h_n) \\
&= \sum_{a\in\mathcal{A}} \mathbb{P}(A_n = a \mid H_n = h_n)\mathbb{P}(S_{n+1} = s' \mid A_n = a, H_n = h_n) \\
&= \sum_{a\in\mathcal{A}} \mathbb{P}(A_n = a \mid S_n = s, S_{n-1} = \tilde{s})\mathbb{P}(S_{n+1} = s' \mid A_n = a, S_n = s) \\
&= \sum_{a\in\mathcal{A}} \pi(a \mid s, \tilde{s})P(s' \mid s, a) \\
&\overset{(a)}{=} \sum_{a\in\mathcal{A}} \mathbb{P}(A_n = a \mid S_n = s, S_{n-1} = \tilde{s})\mathbb{P}(S_{n+1} = s' \mid A_n = a, S_n = s, S_{n-1} = \tilde{s}) \\
&= \sum_{a\in\mathcal{A}} \mathbb{P}(S_{n+1} = s', A_n = a \mid S_n = s, S_{n-1} = \tilde{s}) \\
&= \mathbb{P}(S_{n+1} = s' \mid S_n = s, S_{n-1} = \tilde{s})
\end{aligned}$$

Note that (a) follows from the first-order Markov property of the MDP: conditional on (S_n, A_n), the next state S_{n+1} is independent of the preceding state history.

Therefore, the required transition kernel is

$$\boxed{P^\pi(s' \mid s, \tilde{s}) = \sum_{a\in\mathcal{A}} \pi(a \mid s, \tilde{s})P(s' \mid s, a).}$$

This expression depends on the state history only through (S_n, S_{n-1}), so $(S_n)_{n\geq 0}$ is a second-order Markov chain. □

(b) **General k-th order policy.**

Suppose now that the policy depends on the last k states:

$$\mathbb{P}(A_n = a \mid S_{0:n}, A_{0:n-1}) = \pi(a \mid S_{n-k+1:n}).$$

Show that the state process is an k-th order Markov chain. Derive an expression for its transition kernel in terms of P and π.

Solution. The proof is left as an exercise. It follows exactly the same pattern as part (a), with the last two states replaced by the last k states and with the additional conditioning on the current action A_n. □

2 Action-Value Function

Consider a discounted finite Markov Decision Process $(\mathcal{S}, \mathcal{A}, \mathcal{R}, P, \lambda)$ with discount factor $\lambda \in [0, 1)$. Throughout, $\mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ is equipped with the supremum norm $\|Q\|_\infty := \max_{(s,a)\in\mathcal{S}\times\mathcal{A}} |Q(s,a)|$, under which it is a complete metric space.

In the lectures, the value function v^π of a policy π was introduced. In this question, we develop the analogous state–action value function and its corresponding Bellman operators.

For a policy $\pi \in \Pi^{HR}$, define the state–action value function $Q^\pi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ by

$$Q^\pi(s,a) := \mathbb{E}^\pi_{s,a}\left[\sum_{n=0}^{\infty} \lambda^n r(S_n, A_n)\right],$$

where $\mathbb{E}^\pi_{s,a}$ denotes expectation when the process starts in $S_0 = s$, takes action $A_0 = a$ at time 0, and follows π from time 1 onward.

(a) **Bellman equation for Q^π.**

Let $\pi = \{\phi, \phi, \ldots\}$ be a **stationary** randomized policy for some $\phi : \mathcal{S} \to \Delta(\mathcal{A})$. Write down $Q^\pi(s,a)$ as a recursive expression (Bellman equation).

Solution. Separate the reward at time 0 and condition on (S_1, A_1):

$$\begin{aligned}
Q^\pi(s,a) &= r(s,a) + \mathbb{E}^\pi_{s,a}[\sum_{n=1}^{\infty} \lambda^n r(S_n, A_n)] \\
&= r(s,a) + \mathbb{E}^\pi\left[\mathbb{E}^\pi\left[\sum_{n=1}^{\infty} \lambda^n r(S_n, A_n)\middle| S_1, A_1\right]\right] \\
&= r(s,a) + \sum_{s'\in\mathcal{S}}\sum_{a'\in\mathcal{A}} \mathbb{E}^\pi\left[\sum_{n=1}^{\infty} \lambda^n r(S_n, A_n)\middle| S_1 = s', A_1 = a'\right] \mathbb{P}(S_1 = s', A_1 = a' \mid S_0 = s, A_0 = a) \\
&= r(s,a) + \sum_{s'\in\mathcal{S}}\sum_{a'\in\mathcal{A}} \lambda \underbrace{\mathbb{E}^\pi\left[\sum_{m=0}^{\infty} \lambda^m r(S_{m+1}, A_{m+1})\middle| S_1 = s', A_1 = a'\right]}_{Q^\pi(s',a')} P(s' \mid s,a)\phi(a' \mid s') \\
&= r(s,a) + \lambda \sum_{s'\in\mathcal{S}} P(s' \mid s,a) \sum_{a'\in\mathcal{A}} \phi(a' \mid s') Q^\pi(s',a'). \qquad (1)
\end{aligned}$$

This is the Bellman equation for Q^π. □

(b) Q-**Operator.**

Let $\pi = \{\phi, \phi, ...\}$ be a stationary randomized policy as in (a). Define the operator $L_\pi^Q : \mathbb{R}^{|\mathcal{S}||\mathcal{A}|} \to \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ by

$$(L_\pi^Q \mathbf{Q})(s,a) := r(s,a) + \lambda \sum_{s' \in \mathcal{S}} P(s' \mid s,a) \sum_{a' \in \mathcal{A}} \phi(a' \mid s')\, Q(s',a'). \quad (2)$$

[1] Write L_π^Q in vector form. Show that it is a contraction mapping with modulus λ with respect to $\|\cdot\|_\infty$.

Using the Banach Fixed Point Theorem, conclude that L_π^Q has a unique fixed point.

1. In the question document shared earlier, the Q-Bellman operator L_π^Q incorrectly defined. The definition has been corrected here so that L_π^Q is the standard Bellman operator, with Q^π as its unique fixed point.

Solution. Define the state–action transition matrix

$$P_\pi^{SA}\Big((s,a),(s',a')\Big) := P(s' \mid s,a)\phi(a' \mid s').$$

Then(2) can be written as

$$(L_\pi^Q \mathbf{Q})(s,a) = r(s,a) + \lambda \sum_{(s',a') \in \mathcal{S}\times\mathcal{A}} P_\pi^{SA}\Big((s,a),(s',a')\Big) Q(s',a').$$

Viewing $\mathbf{Q}$ and $\mathbf{r}$ as vectors indexed by state–action pairs, we obtain

$$\boxed{L_\pi^Q \mathbf{Q} = \mathbf{r} + \lambda P_\pi^{SA} \mathbf{Q}.}$$

In particular, evaluating this operator at $\mathbf{Q} = \mathbf{Q}^\pi$ recovers the Bellman equation (1). Hence,

$$\boxed{\mathbf{Q}^\pi = L_\pi^Q \mathbf{Q}^\pi.}$$

The matrix P_π^{SA} is row-stochastic. Therefore, for any $\mathbf{Q}_1, \mathbf{Q}_2 \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$,

$$\begin{aligned} \|L_\pi^Q \mathbf{Q}_1 - L_\pi^Q \mathbf{Q}_2\|_\infty &= \lambda \|P_\pi^{SA}(\mathbf{Q}_1 - \mathbf{Q}_2)\|_\infty \\ &\le \lambda \|\mathbf{Q}_1 - \mathbf{Q}_2\|_\infty. \end{aligned}$$

Because $0 \le \lambda < 1$, L_π^Q is a contraction on the complete space $\mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$. The Banach Fixed Point Theorem therefore gives a unique fixed point. By the above, this fixed point is Q^π. □

(c) **Optimal state–action value function.**

Define the optimal state–action value function by

$$Q^*_{\infty,\lambda}(s,a) := \sup_{\pi \in \Pi^{HR}} Q^\pi(s,a).$$

Consider the operator L^Q defined by

$$(L^Q \mathbf{Q})(s,a) := r(s,a) + \lambda \sum_{s' \in \mathcal{S}} P(s' \mid s,a) \max_{a' \in \mathcal{A}} Q(s',a'). \quad (3)$$

(i) Show that $Q^*_{\infty,\lambda}$ satisfies the fixed-point equation

$$Q^*_{\infty,\lambda} = L^Q Q^*_{\infty,\lambda}.$$

Solution. After the prescribed initial action a, choose an optimal continuation from the next state. The principle of optimality yields

$$\begin{aligned} Q^*_{\infty,\lambda}(s,a) &= r(s,a) + \lambda \sum_{s' \in \mathcal{S}} P(s' \mid s,a) \max_{a' \in \mathcal{A}} Q^*_{\infty,\lambda}(s',a') \qquad (4) \\ &= (L^Q Q^*_{\infty,\lambda})(s,a). \end{aligned}$$

Because this holds for every (s,a), $Q^*_{\infty,\lambda} = L^Q Q^*_{\infty,\lambda}$. □

(ii) Show that L^Q is a contraction mapping with modulus λ with respect to $\|\cdot\|_\infty$. Hence, using the Banach Fixed Point Theorem, conclude that $Q^*_{\infty,\lambda}$ is the unique fixed point of L^Q.

Solution. For finite vectors $(x_a)_a$ and $(y_a)_a$,

$$\left|\max_a x_a - \max_a y_a\right| \le \max_a |x_a - y_a|.$$

Consequently, for every $(s,a) \in \mathcal{S} \times \mathcal{A}$,

$$\begin{aligned} |(L^Q Q_1)(s,a) - (L^Q Q_2)(s,a)| &\overset{(3)}{=} \lambda \Big| \sum_{s' \in \mathcal{S}} P(s' \mid s,a)(\max_{a'} Q_1(s',a') - \max_{a'} Q_2(s',a'))\Big| \\ &\le \lambda \sum_{s'} P(s' \mid s,a) \max_{a'} |Q_1(s',a') - Q_2(s',a')| \\ &\le \lambda \|\mathbf{Q_1} - \mathbf{Q_2}\|_\infty. \end{aligned}$$

Taking the maximum over (s,a) proves

$$\|L^Q \mathbf{Q_1} - L^Q \mathbf{Q_2}\|_\infty \le \lambda \|\mathbf{Q_1} - \mathbf{Q_2}\|_\infty.$$

Thus L^Q is a contraction. Banach's theorem gives a unique fixed point, which is $Q^*_{\infty,\lambda}$ by part (i). □

(d) **Connection between $v^*_{\infty,\lambda}$ and $Q^*_{\infty,\lambda}$.**

Show that

$$v^*_{\infty,\lambda}(s) = \max_{a \in \mathcal{A}} Q^*_{\infty,\lambda}(s,a).$$

Furthermore, show that

$$Q^*_{\infty,\lambda}(s,a) = r(s,a) + \lambda \sum_{s' \in \mathcal{S}} P(s' \mid s,a) v^*_{\infty,\lambda}(s').$$

Solution. We first prove the two inequalities separately.

Fix $a \in \mathcal{A}$ and a continuation policy π. Define a policy $\tilde{\pi}$ that selects a with probability one at time 0 and follows π from time 1 onward. By the definition of the action-value function,

$$Q^\pi(s,a) = v^{\tilde{\pi}}_{\infty,\lambda}(s) \le v^*_{\infty,\lambda}(s).$$

Taking the supremum over all continuation policies π, and then the maximum over a, gives

$$\max_{a\in\mathcal{A}} Q^*_{\infty,\lambda}(s,a) \le v^*_{\infty,\lambda}(s).$$

For the reverse inequality, fix an arbitrary policy π, thus $\pi_0(a \mid s)$ denotes its initial action distribution. After conditioning on $A_0 = a$, let $\pi_{1:}^{s,a}$ denote the resulting continuation policy from time 1 onward. Then

$$\begin{aligned} v^{\pi}_{\infty,\lambda}(s) &= \sum_{a\in\mathcal{A}} \pi_0(a \mid s) Q^{\pi_{1:}^{s,a}}(s,a) \\ &\le \sum_{a\in\mathcal{A}} \pi_0(a \mid s) Q^*_{\infty,\lambda}(s,a) \\ &\le \max_{a\in\mathcal{A}} Q^*_{\infty,\lambda}(s,a). \end{aligned}$$

Taking the supremum over all policies π on the LHS yields

$$v^*_{\infty,\lambda}(s) \le \max_{a\in\mathcal{A}} Q^*_{\infty,\lambda}(s,a).$$

Combining the two inequalities proves

$$\boxed{v^*_{\infty,\lambda}(s) = \max_{a\in\mathcal{A}} Q^*_{\infty,\lambda}(s,a)}.$$

Substituting this identity into (4) gives

$$\begin{aligned} Q^*_{\infty,\lambda}(s,a) &= r(s,a) + \lambda \sum_{s'} P(s' \mid s,a) \max_{a'} Q^*_{\infty,\lambda}(s',a') \\ &= r(s,a) + \lambda \sum_{s'} P(s' \mid s,a) v^*_{\infty,\lambda}(s'). \end{aligned}$$

Thus, either optimal value function determines the other. □

(e) **Optimal action and optimal policy.**

Using the optimal state–action value function, define

$$\pi^*(s) \in \arg\max_{a\in\mathcal{A}} Q^*_{\infty,\lambda}(s,a).$$

Show that

$$v^*_{\infty,\lambda}(s) = Q^*_{\infty,\lambda}(s,\pi^*(s))$$

and explain why π^* is an optimal policy.

Solution. By the definition of a greedy action and part (d),

$$v^*_{\infty,\lambda}(s) = \max_{a\in\mathcal{A}} Q^*_{\infty,\lambda}(s,a) = Q^*_{\infty,\lambda}(s,\pi^*(s)).$$

Let L_{π^*} denote the Bellman operator for the deterministic stationary policy π^*. Using the relation between Q^* and v^*,

$$\begin{aligned} (L_{\pi^*} v^*_{\infty,\lambda})(s) &= r(s,\pi^*(s)) + \lambda \sum_{s'} P(s' \mid s,\pi^*(s)) v^*_{\infty,\lambda}(s') \\ &= Q^*_{\infty,\lambda}(s,\pi^*(s)) \\ &= v^*_{\infty,\lambda}(s). \end{aligned}$$

Thus $v^*_{\infty,\lambda}$ is a fixed point of L_{π^*}. Since this policy Bellman operator has a unique fixed point, $v^{\pi^*}_{\infty,\lambda} = v^*_{\infty,\lambda}$. Therefore π^* is optimal. □